

AMROHA POLICE CYBER SECURITY INTERNSHIP PROGRAM 2026

INTERNSHIP PROJECT REPORT

CRIMEVISION AI

Intelligent Conversational Crime Analytics &
Investigation Support Platform

Submitted By

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Submitted To

Amroha Police Cyber Cell

Amroha Police Cyber Security Internship Program 2026

ABSTRACT

CrimeVision AI is an advanced crime intelligence platform designed to support investigators, analysts, supervisors, and policymakers through AI-driven crime analytics and conversational intelligence.

The platform combines natural language processing, criminal network analysis, crime forecasting, offender profiling, financial crime intelligence, explainable artificial intelligence, and secure governance mechanisms into a unified law enforcement dashboard.

The system aims to transform traditional policing into intelligence-led policing by enabling proactive crime prevention, faster investigations, and data-driven decision-making.

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1. INTRODUCTION

Modern law enforcement agencies generate vast amounts of information through FIRs, investigation records, cybercrime complaints, criminal databases, and intelligence reports. Extracting meaningful insights from this data remains a significant challenge.

CrimeVision AI addresses this challenge by providing an intelligent platform capable of:

- Conversational crime intelligence
- Criminal relationship discovery
- Financial crime analysis
- Crime trend analytics
- Offender profiling
- Predictive crime forecasting
- Explainable AI-driven decision support

2. PROBLEM STATEMENT

Existing crime management systems primarily focus on record keeping and manual search operations. CrimeVision AI is designed to overcome the following challenges using advanced artificial intelligence technologies:

- Fragmented information systems across departments and jurisdictions
- Limited analytical capabilities for pattern detection
- Lack of predictive intelligence for proactive policing
- Difficulty identifying organized criminal networks
- Time-consuming manual investigations
- Limited decision support tools for investigators and supervisors

3. OBJECTIVES

1. Build a conversational crime intelligence platform.
2. Enable natural language crime database interaction.
3. Detect criminal networks and associations.
4. Perform offender profiling and risk assessment.
5. Analyze financial crime transactions.
6. Forecast crime hotspots and emerging threats.
7. Provide explainable AI recommendations.
8. Ensure secure role-based access management.

4. LITERATURE REVIEW

The project is grounded in concepts and research derived from the following domains:

- Artificial Intelligence in Policing
- Predictive Crime Analytics
- Criminal Network Analysis
- Graph Database Technologies
- Cyber Threat Intelligence Systems
- Explainable Artificial Intelligence (XAI)
- Crime and Criminal Tracking Network Systems (CCTNS)

5. PROPOSED SOLUTION

CrimeVision AI delivers an integrated suite of intelligence capabilities:

Conversational Crime Assistant

Natural language interaction with crime databases enabling investigators to query records without technical expertise.

Criminal Network Analysis

Relationship mapping among suspects, victims, locations, vehicles, and financial accounts using graph-based analytics.

Crime Analytics

Pattern identification and hotspot detection across jurisdictions with temporal analysis.

Offender Profiling

Behavioral and criminological analysis with recidivism risk scoring.

Financial Intelligence

Money trail tracking and suspicious transaction detection for anti-money laundering investigations.

Crime Forecasting

Predictive intelligence and early warning alerts using LSTM-based time series models.

Explainable AI

Transparent recommendations with evidence trails and confidence scoring.

Governance

Secure role-based access control, audit logging, and user activity tracking.

6. SYSTEM ARCHITECTURE

Data Sources

- FIR Records
- Criminal Records
- Cyber Crime Reports
- Financial Transactions
- Intelligence Databases
- Demographic Data

Processing Layer

- AI Analytics Engine
- NLP Engine
- Risk Assessment Models
- Criminal Network Analysis

Presentation Layer

- Dashboard
- Chat Assistant
- Maps
- Graph Analytics
- Investigation Support Tools

7. TECHNOLOGY STACK

Category	Technologies
Frontend	HTML5, CSS3, JavaScript ES6+
Visualization	Chart.js, Leaflet.js, Cytoscape.js
Reporting	html2pdf.js
UI Design	Glassmorphism UI, Responsive Dashboard, SPA Architecture

8. IMPLEMENTATION METHODOLOGY

The project follows a modular single-page application (SPA) architecture. The file structure is as follows:

index.html	Application shell and layout
styles.css	UI design system and responsive styling
mockData.js	Crime records and intelligence datasets
utils.js	PDF export, voice support, audit logging
components.js	Dashboard modules and renderers
app.js	Routing, RBAC engine, and application logic

9. MODULE DESCRIPTIONS

Module 1 – Secure Access Portal

Provides encrypted access for authorized law enforcement personnel with multi-factor authentication support.

- Credential-based authentication
- Biometric simulation
- Role selection for RBAC enforcement

Module 2 – Dashboard Overview

Central command view providing real-time situational awareness.

- Active cases counter
- Critical alerts panel
- Real-time incident geoplotter
- Security audit ledger with live monitoring

Module 3 – AI Intelligence Copilot

Conversational AI assistant for natural language crime intelligence queries.

- Natural language queries
- Voice commands via speech recognition
- Multi-language interaction (English, Hindi, Kannada)
- PDF export of intelligence reports

Module 4 – Crime Analytics

Advanced visualization and pattern analysis of crime data.

- Interactive crime heatmaps
- Crime category share (polar graph)
- Monthly incident volumetric trends
- Seasonal profile correlations

Module 5 – Criminal Network Analysis

Graph-based relationship discovery using Neo4j-schema integrated visualization.

- Organized criminal relationship graphing
- Gang cluster detection
- Network node filtering by type (accused, victim, bank, phone, vehicle)

Module 6 – Offender Profiling

Comprehensive dossier management with behavioral risk assessment.

- Behavioral analysis with sociopathy and recidivism scoring
- Criminal history timeline
- Investigator recommendations

Module 7 – Financial Crime Intelligence

Money trail visualization and transaction audit capabilities.

- Suspicious transaction money trail flow diagrams
- Transaction audit ledger
- Layering and laundering status flagging

Module 8 – Crime Forecasting

Predictive intelligence using LSTM-based auto-regressive forecasting models.

- Predictive crime hotspot simulation maps
- High risk jurisdiction prediction
- Early warning system alert logs

Module 9 – Explainable AI

Transparent AI recommendations with auditable reasoning chains.

- Confidence scoring
- Evidence trails
- Transparent recommendations

10. USER INTERFACE SCREENS

Figure 1 – CrimeVision AI Landing Portal

The landing portal presents the system identity and directs authorized personnel to the encrypted authentication gateway.

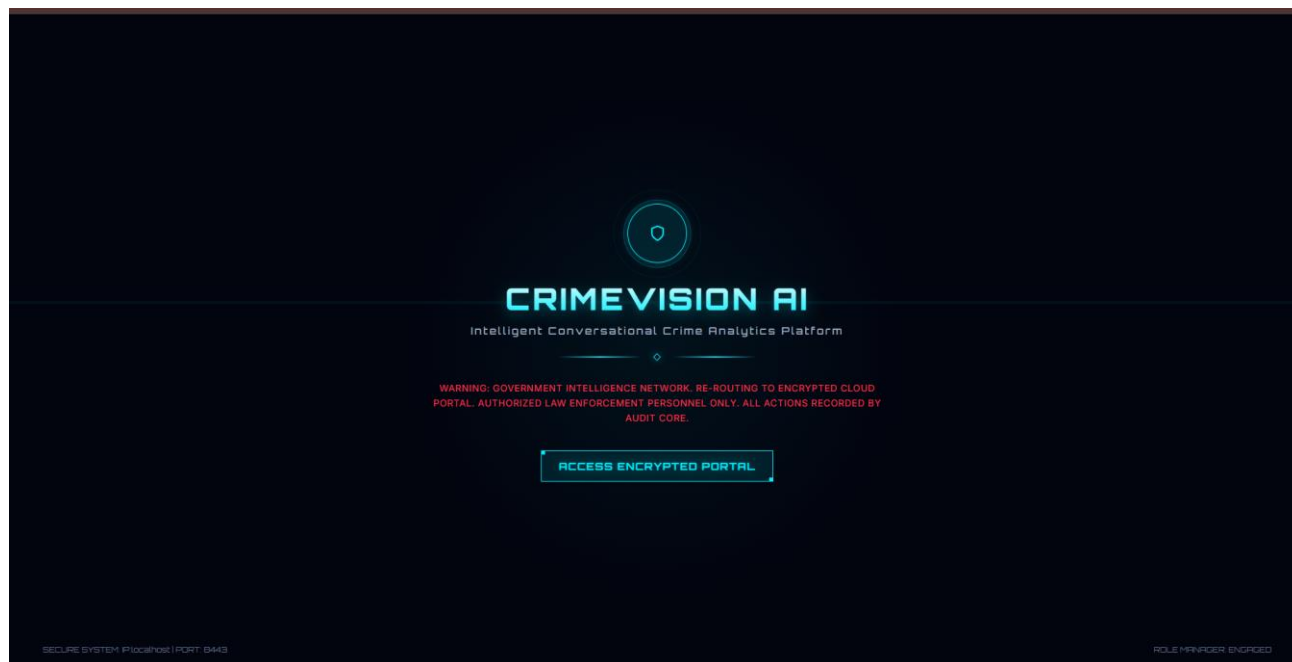


Figure 1: CrimeVision AI Landing Portal – Intelligent Conversational Crime Analytics Platform

Figure 2 – Secure Authentication Interface

The authentication screen supports credential entry, secure keypass, role selection, and biometric simulation for multi-factor access control.

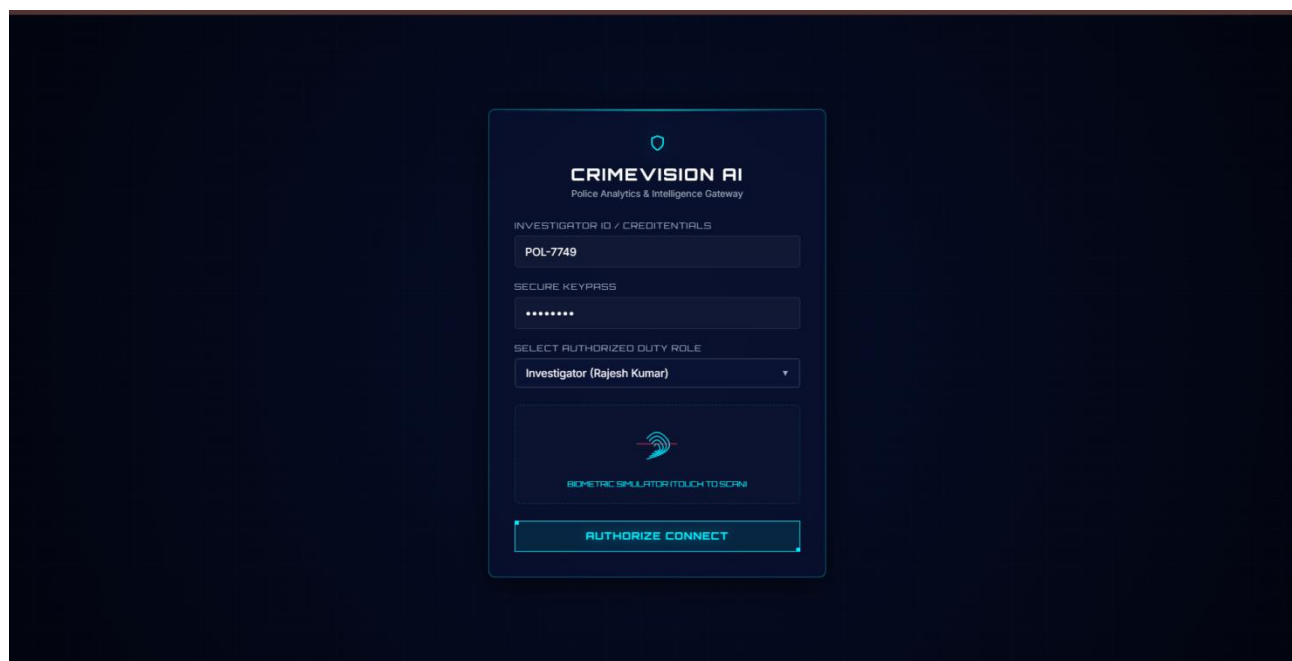


Figure 2: Secure Authentication Interface – Role-Based Biometric Login Portal

Figure 3 – Dashboard Overview Center

The Dashboard Module Control provides a real-time situational overview including active cases, critical alerts, solved case rates, live geoplotted, and security audit ledger.

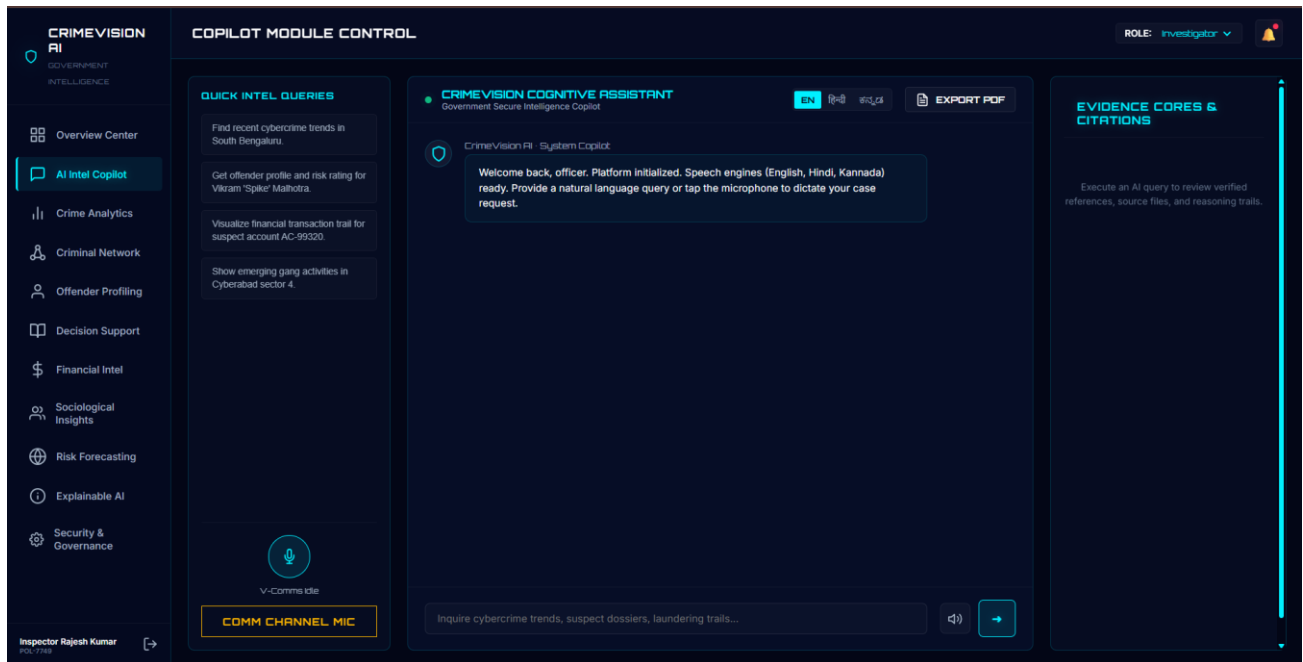


Figure 3: Dashboard Overview Center – Real-Time Police Intelligence Command View

Figure 4 – AI Intelligence Copilot

The Copilot Module Control enables natural language interaction with crime databases, voice commands, multi-language support, and evidence citation export.

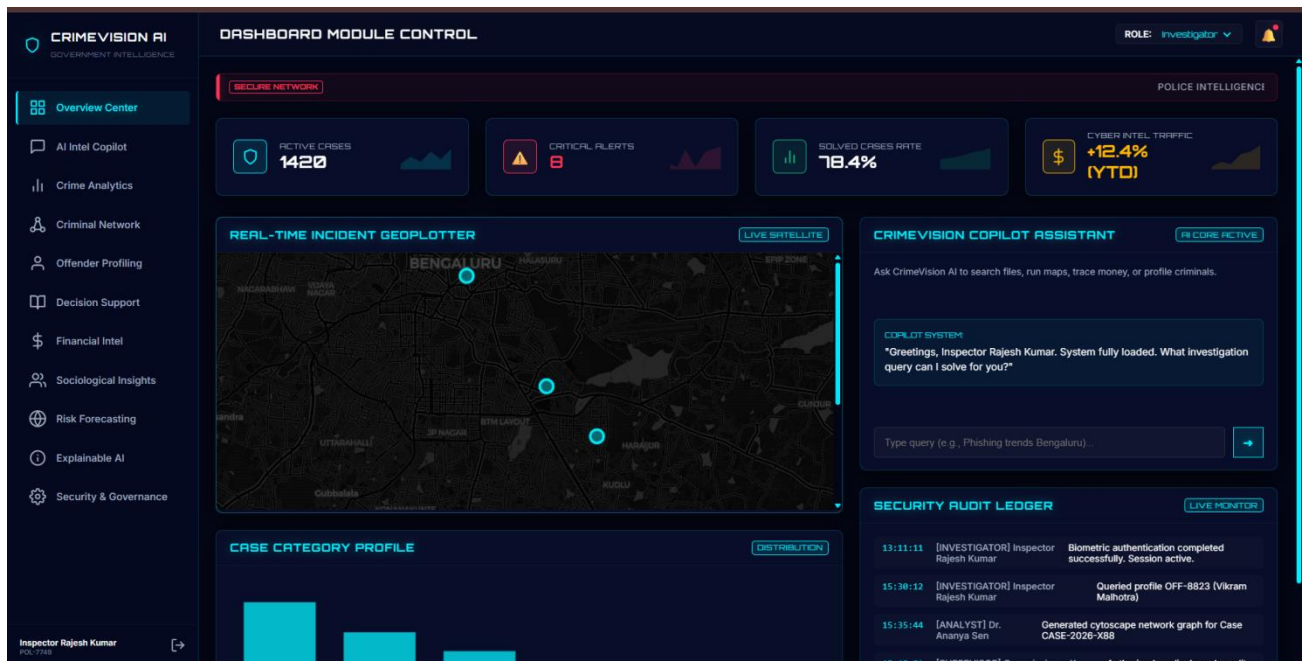


Figure 4: AI Intelligence Copilot – Government Secure Intelligence Chat Interface

Figure 5 – Crime Analytics Dashboard

The Analytics Module Control displays interactive incident heatmaps, crime category distribution, monthly volumetric trends, and seasonal correlation analysis for Bengaluru jurisdiction.

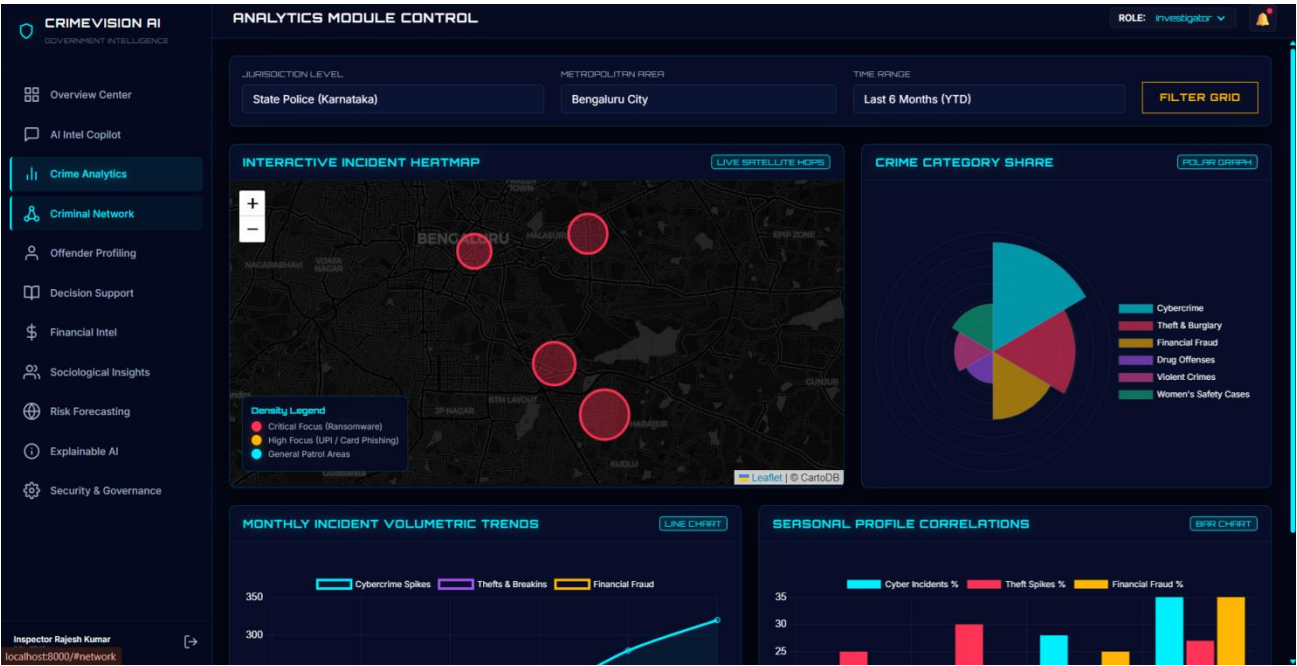


Figure 5: Crime Analytics Dashboard – Interactive Incident Heatmap and Category Analysis

Figure 6 – Criminal Network Analysis Dashboard

The Network Module Control renders an organized criminal relationship graph with Neo4j integration, supporting gang cluster detection and multi-entity relationship mapping.

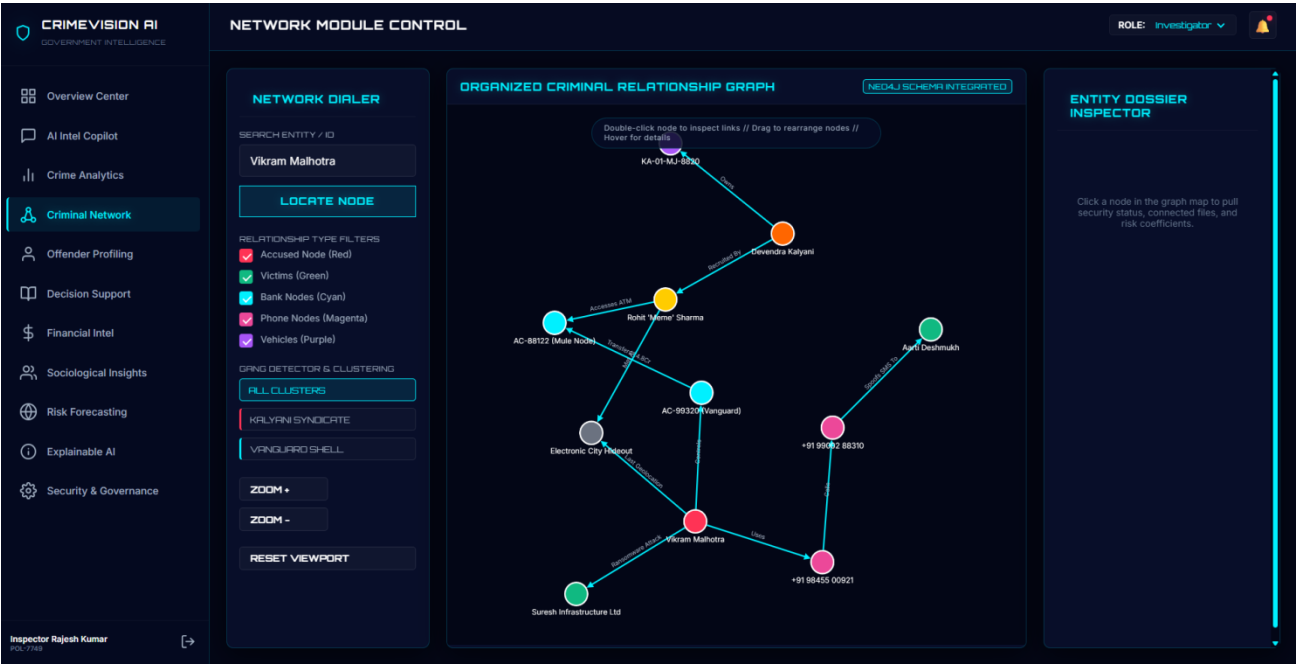


Figure 6: Criminal Network Analysis – Organized Criminal Relationship Graph Visualization

Figure 7 – Offender Profiling Dashboard

The Offenders Module Control presents detailed dossiers with behavioral risk assessment scores, modus operandi signature analysis, and full criminal log timelines.

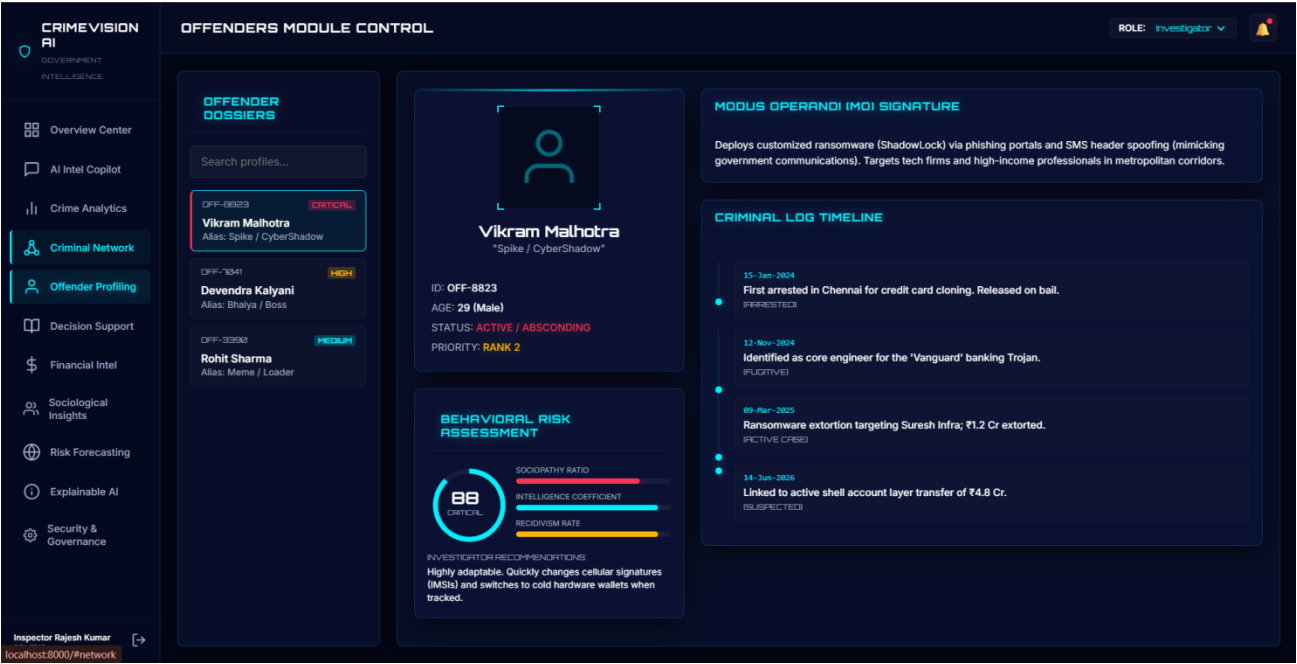


Figure 7: Offender Profiling Dashboard – Behavioral Risk Assessment and Criminal Dossier

Figure 8 – Financial Intelligence Dashboard

The Financial Module Control visualizes suspicious transaction money trail flows and provides a transaction audit ledger with laundering status classification.

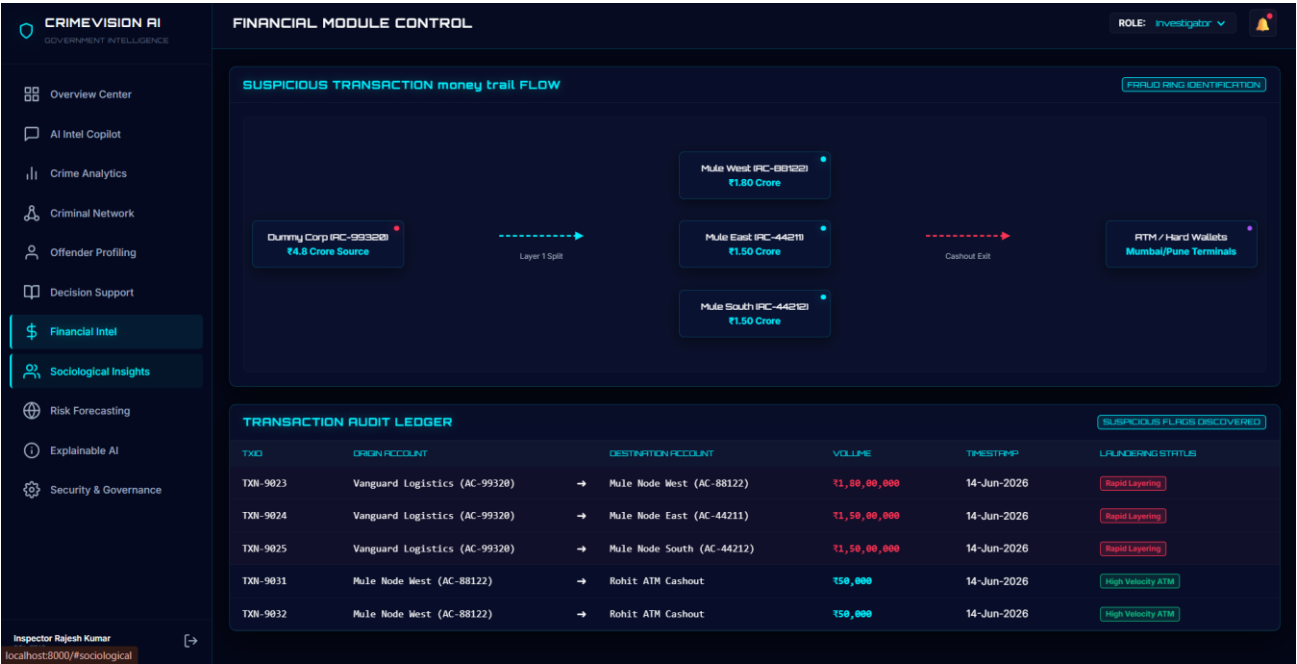


Figure 8: Financial Crime Intelligence – Suspicious Transaction Money Trail Flow Analysis

Figure 9 – Crime Forecasting Dashboard

The Forecasting Module Control presents predictive crime hotspot simulations, high-risk jurisdiction predictions for the next 30 days, early warning alerts, and LSTM-based monthly trend forecasts.

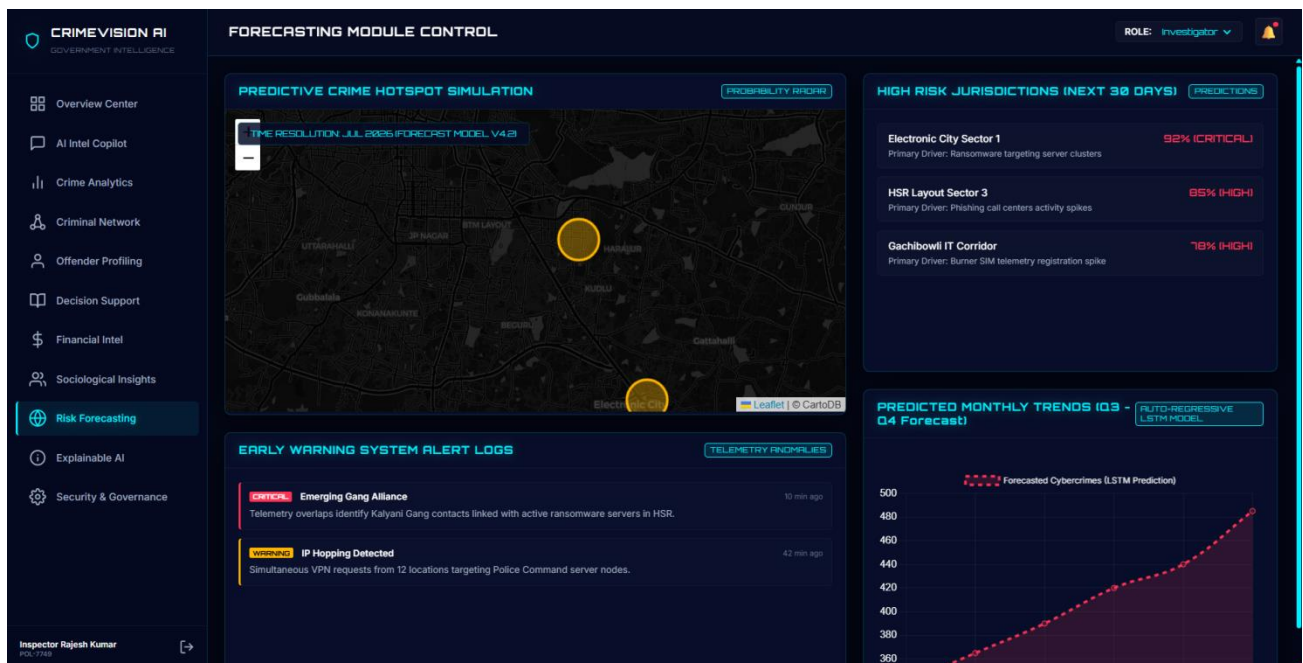


Figure 9: Crime Forecasting Dashboard – Predictive Hotspot Simulation and Risk Intelligence

11. SECURITY FRAMEWORK

The platform implements a comprehensive multi-layered security architecture:

- Role-Based Access Control (RBAC) – Granular permission management by user role
- Audit Logging – Immutable timestamped record of all user actions
- Secure Authentication – Multi-factor login with biometric simulation
- User Activity Tracking – Real-time session monitoring
- Data Governance Policies – Controlled access to sensitive intelligence data
- Access Monitoring – Live audit ledger visible on the dashboard

12. TESTING AND VALIDATION

Comprehensive testing was conducted across all major modules with the following results:

Test Area	Result
Navigation Testing	Pass
Dashboard Functionality Testing	Pass
Crime Analytics Verification	Pass
Network Visualization Testing	Pass
Voice Interaction Testing	Pass
PDF Export Testing	Pass
Role-Based Access Testing	Pass

All major modules confirmed successful operation during testing and validation phases.

13. RESULTS AND OUTCOMES

The CrimeVision AI project successfully demonstrates:

- AI-powered crime intelligence with real-time conversational querying
- Criminal relationship discovery through graph-based network analysis
- Crime pattern analysis and interactive heatmap visualization
- Financial crime investigation support with money trail mapping
- Predictive policing capabilities using LSTM forecasting models
- Explainable AI recommendations with confidence scoring and evidence trails
- Secure governance framework with RBAC and full audit logging

14. FUTURE SCOPE

Future enhancements planned for CrimeVision AI include:

- CCTNS Integration – Connectivity with the national crime database
- Real-time Crime Feeds – Live data ingestion from FIR and incident systems
- Mobile Application Development – iOS and Android companion apps
- Face Recognition Integration – Biometric suspect identification
- Social Media Intelligence Analysis – OSINT capabilities
- Advanced Deep Learning Models – Improved accuracy and threat prediction
- Multi-language Expansion – Support for additional Indian regional languages

15. CONCLUSION

CrimeVision AI demonstrates how Artificial Intelligence, Cyber Security, Crime Analytics, and Data Science can significantly enhance law enforcement capabilities.

This project highlights the potential of emerging technologies to improve public safety, investigation efficiency, and proactive crime prevention, representing a meaningful contribution to the field of AI-assisted law enforcement.

16. REFERENCES

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